

Undergraduate Committee Report: Faculty Meeting, September 7, 2018

5. Fall Semester Enrollment:

	<u>2017</u>	<u>2018 (as of 9/4/18)</u>
o Total:	279	261
o Freshmen:	36	30
o Transfers:	36	24
o UIC Global:	2	18
o Total w/o UICGlobal:	277	243

Out of 277 students in 2017, 70 graduated to give us 207 students. We lost 18 continuing students to bring the total to 189 students. We added 54 freshmen/transfers giving us the 243 student total without UIC Global. Adding the UIC Global gives us 261 total.

6. Fall Recruiting Efforts

- UIC Open House SATURDAY SEPTEMBER 22th 9am-12noon
 - o ChE presentation (Zdunek)
 - o CHE demonstrations (Bilgin, Mehraeen, Sharma, Berry, Kim)
 - o Location: ERF 1043
- CHE Open House for Undeclared Engineering students, August 23, 2018
- Other (Naperville HS recruiting, COD career night, Wheeling HS)

7. UG Activities this Semester

- 4th Annual Industry Day Event Friday October 19th 3pm-6:30pm (Networking added)
- ChE Club (new initiative) Periodic fun events throughout the semester

8. Curriculum Day, Tuesday October 16th, 9am-12noon

- Current curriculum and discussion on improvements. Led by Zdunek
- Best Practices in Teaching: Wedgewood
- Best Practices in mentoring UGs for research: Sharma
- Teaching Methods: Bilgin
- Lessons from Chem-E-Car Experience: Singh
- Lunch + Discussion on Improving student experience: Led by Berry and Nitsche.

9. ABET --- Fall 2020

- Faculty Preparatory Meeting this semester
 - Courses output needed/procedure
- Review PEO's
- New ABET Student Learning Outcomes
 - Mapping to current courses
 - Map Old (2014) to New

see attached AZ

AlCHE Mtg: Oct 28 - Nov 2 week 10!

change

Advising Week 9

2.A.3. Mission statement of the Department Chemical Engineering

“The mission of the Department of Chemical Engineering is to serve society through excellence in education, research, and public service. We provide for our students an education in chemical engineering, and we aspire to instill in them the attitudes, values, and vision that will prepare them for lifetimes of continued learning and leadership in their chosen careers. Through scholarship, the Department strives to generate new knowledge and technology for the benefit of the State of Illinois, the nation, and beyond.”

The departmental mission statement is available online to students and all other College constituencies at the following URL.

<http://www.che.uic.edu/CHE/Welcome>

B. Program Educational Objectives

PEO	Category	Description
1	Depth	In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.
2	Breadth	Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3	Professionalism and Service	In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.
4	Life-Long Learning and Leadership	Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.

The Program Educational Objectives are prominently listed in the “Objectives and Outcomes” link on the home page of the departmental website: www.che.uic.edu. Additionally, the PEOs are included in all undergraduate course syllabi, and class time is devoted to explaining how the knowledge, skills and principles developed by the course in question contribute to the professional capacities represented by the PEOs.

UIC Student Outcomes, ABET 2014

Student Outcomes 2014

New SO

Previous

SO	Description	ABET	PEO
1	Ability to apply knowledge of mathematics, science and engineering	a	1
2	Ability to identify, formulate and solve engineering problems	e	1
3	Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice	k	1
4	Ability to design experiments	b	1
5	Ability to conduct experiments	b	1
6	Ability to analyze and interpret data	b	1
7	Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability	c	1
8	Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context	h	2
9	Knowledge of contemporary issues	j	2
10	Recognition of the need for and ability to engage in life-long learning	i	4
11	Ability to function on multidisciplinary teams	d	3
12	Ability to communicate effectively	g	3
13	Understanding of professional and ethical responsibilities	f	3

1

6

2

4

7

5

3

4



New ABET Student Outcomes

Criterion 3. Student Outcomes

The program must have documented student outcomes that support the program educational objectives. Attainment of these outcomes prepares graduates to enter the professional practice of engineering. Student outcomes are outcomes (1) through (7), plus any additional outcomes that may be articulated by the program.

- ✓ (1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
- ✓ (2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
- ✓ (3) an ability to communicate effectively with a range of audiences
- ✓ (4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
- ✓ (5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
- ✓ (6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
- ✓ (7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.



DATE: October 20, 2017
TO: Engineering Area Delegation
FROM: Ann Kenimer, Chair, Engineering Accreditation Commission
SUBJECT: Proposed Revised to the *Criteria for Accrediting Engineering Programs* General Criteria Introduction, Criterion 3. Student Outcomes, and Criterion 5. Curriculum – Second Reading

RECOMMENDED MOTION:

The Engineering Area Delegation approves the attached *Proposed Program Criteria for Accrediting Engineering Programs General Criteria Introduction, Criterion 3. Student Outcomes, and Criterion 5. Curriculum* at the second reading for implementation by the Engineering Accreditation Commission (EAC) in the 2019-20 accreditation cycle.

BACKGROUND:

The Engineering Accreditation Commission requests that the following changes in the Engineering General Criteria and supporting text be approved by the Engineering Area Delegation for implementation in the 2019-20 accreditation cycle.



PROPOSED REVISION
TO
Criteria for Accrediting Engineering Programs

Introduction

These criteria apply to all accredited engineering programs. Furthermore, these criteria are intended to foster the systematic pursuit of improvement in the quality of engineering education that satisfies the needs of its constituencies in a dynamic and competitive environment. It is the responsibility of the institution seeking accreditation of an engineering program to demonstrate clearly that the program meets the following criteria.

Definitions

The Engineering Accreditation Commission of ABET recognizes that its constituents may consider certain terms to have certain meanings; however, it is necessary for the Engineering Accreditation Commission to have consistent terminology. Thus, the Engineering Accreditation Commission will use the following definitions in applying the criteria:

Basic Science – Basic sciences are disciplines focused on knowledge or understanding of the fundamental aspects of natural phenomena. Basic sciences consist of chemistry and physics and other natural sciences including life, earth, and space sciences.

College-Level Mathematics – College-level mathematics consists of mathematics that requires a degree of mathematical sophistication at least equivalent to that of introductory calculus. For illustrative purposes, some examples of college-level mathematics include calculus, differential equations, probability, statistics, linear algebra, and discrete mathematics.



Complex Engineering Problems - Complex engineering problems include one or more of the following characteristics: involving wide-ranging or conflicting technical issues, having no obvious solution, addressing problems not encompassed by current standards and codes, involving diverse groups of stakeholders, including many component parts or sub-problems, involving multiple disciplines, or having significant consequences in a range of contexts.

Engineering Design – Engineering design is a process of devising a system, component, or process to meet desired needs and specifications within constraints. It is an iterative, creative, decision-making process in which the basic sciences, mathematics, and engineering sciences are applied to convert resources into solutions. Engineering design involves identifying opportunities, developing requirements, performing analysis and synthesis, generating multiple solutions, evaluating solutions against requirements, considering risks, and making trade-offs, for the purpose of obtaining a high-quality solution under the given circumstances. For illustrative purposes only, examples of possible constraints include accessibility, aesthetics, codes, constructability, cost, ergonomics, extensibility, functionality, interoperability, legal considerations, maintainability, manufacturability, marketability, policy, regulations, schedule, standards, sustainability, or usability.

Engineering Science – Engineering sciences are based on mathematics and basic sciences but carry knowledge further toward creative application needed to solve engineering problems. These studies provide a bridge between mathematics and basic sciences on the one hand and engineering practice on the other.

Team – A team consists of more than one person working toward a common goal and should include individuals of diverse backgrounds, skills, or perspectives.



Criterion 3. Student Outcomes

The program must have documented student outcomes that support the program educational objectives. Attainment of these outcomes prepares graduates to enter the professional practice of engineering. Student outcomes are outcomes (1) through (7), plus any additional outcomes that may be articulated by the program.

- (1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
- (2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
- (3) an ability to communicate effectively with a range of audiences
- (4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
- (5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
- (6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
- (7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

**Criterion 5. Curriculum**

The curriculum requirements specify subject areas appropriate to engineering but do not prescribe specific courses. The program curriculum must provide adequate content for each area, consistent with the student outcomes and program educational objectives, to ensure that students are prepared to enter the practice of engineering. The curriculum must include:

- (a) a minimum of 30 semester credit hours (or equivalent) of a combination of college-level mathematics and basic sciences with experimental experience appropriate to the program.
- (b) a minimum of 45 semester credit hours (or equivalent) of engineering topics appropriate to the program, consisting of engineering and computer sciences and engineering design, and utilizing modern engineering tools.
- (c) a broad education component that complements the technical content of the curriculum and is consistent with the program educational objectives.
- (d) a culminating major engineering design experience that 1) incorporates appropriate engineering standards and multiple constraints, and 2) is based on the knowledge and skills acquired in earlier course work.